Journal of Electromyography and Kinesiology 20 (2010) 1107–1114

Contents lists available at ScienceDirect

Journal of Electromyography and Kinesiology

journal homepage: www.elsevier.com/locate/jelekin

Distribution of motor unit potential velocities in the biceps brachii muscle of sprinters and endurance athletes during short static contractions at low force levels

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ARTICLE INFO

Article history:
Received 9 July 2009
Received in revised form 19 May 2010
Accepted 19 May 2010

Keywords:
Surface electromyography
Position tasks
Skewness
Muscle fiber conduction velocity

ABSTRACT

In surface electromyography (sEMG), the distribution of motor unit potential (MUP) velocities has been shown to reflect the proportion of faster and slower propagating MUPs. This study investigated whether the distribution of MUP velocities could distinguish between sprinters and endurance athletes in not-specifically trained muscle (biceps brachii). sEMG results were acquired from 15 sprinters and 18 endurance athletes during short static contractions (3.8 s) at three force levels: unloaded, 10% and 20% of maximum voluntary contraction. The features extracted from the sEMG were: the mean muscle conduction velocity (CV) – estimated using the inter-peak latency and the cross-correlation methods, the within-subject skewness of MUP velocities (expressing the relative proportions of faster and slower propagating MUPs), and the within-subject standard deviation of MUP velocities. Sprinters had a higher CV than endurance athletes using both methods. Sprinters also demonstrated a greater proportion of fast propagating MUPs, as indicated by the skewness. Thus, the distribution of MUP velocities was able to demonstrate physiological differences between sprinters and endurance athletes during short contractions at low forces. The findings can be extrapolated to the motor unit level. Since the investigated muscle was not involved in specific training, the differences seem to reflect inherited properties.

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1. Introduction

In a recent study, we introduced a method that used the distribution of motor unit potential (MUP) velocities as a measure of the proportions of faster and slower propagating MUPs in surface electromyography (sEMG) (Klaver-Król et al., 2007). The present study investigates whether the method is able to distinguish between the muscle properties of sprinters and endurance athletes.

It is recognized that the physiological properties of the skeletal muscles of sprinters and endurance athletes differ (Brooks et al., 2005; Gollnick et al., 1972). In simplified terms, the muscles of a sprinter rely more on anaerobic metabolism, and this is attributed to a preponderance of large, fast contracting, fatigue-sensitive type FF (=type IIb) muscle fibers. In contrast, the muscles of an endurance athlete are dominated by the aerobic metabolism associated with smaller, slow contracting, fatigue-resistant type S (=type I) fibers (Buchthal et al., 1973; Burke et al., 1973). Muscle fibers (with their motor units), in a muscle, form a spectrum ranging from the FF to the S types, with many intermediate fibers/units (Kernell, 2006). Since FF fibers, belonging to the larger motor units (MUs), propagate an action potential faster along the membrane than S fibers, belonging to smaller MUs (Andreassen and Arendt-Nielsen, 1987; Buchthal et al., 1973), the mean conduction velocity of a given muscle will depend on the

Abbreviations: MU, motor unit; MUP, motor unit potential; EMG, electromyography; sEMG, surface electromyography; IPL, inter-peak latency method; CC, cross-correlation method; MVC, maximum voluntary contraction; CV, mean muscle conduction velocity; CV-ipl, mean muscle conduction velocity estimated by the inter-peak latency technique; CV-cc, mean muscle conduction velocity estimated by the cross-correlation technique; MUP-V, MUP velocity/peak velocity = propagation velocity of an MUP/peak; Sk, skewness of the within-subject MUP velocities; SD-mup, standard deviation of the within-subject MUP velocities; FF type, fast-twitch/fatigue-sensitive muscle fiber type = fast/glycogenic FG = type IIb; S type, slow-twitch/fatigue-resistant muscle fiber type = slow/oxidative SO = type I.

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doi:10.1016/j.jelekin.2010.05.008

proportions of the fiber/MU types within it. In line with this Sadoyama et al. (1988) have found that the mean muscle conduction velocity (CV) correlates linearly with the area of anaerobic (FF) muscle fibers in the vastus lateralis muscle of sprinters and distance runners.

The CV assessments can be refined by measuring the propagation velocities of individual motor unit potentials (MUPs). One technique is the inter-peak latency method (IPL) proposed by Lange et al. (2002) in which the MUP velocities are calculated from the time intervals between paired MUPs of two differential sEMG signals obtained parallel to the muscle fibers. In a previous study, we quantified the MUP velocities estimated by the IPL method using two statistical distribution parameters: the within-subject skewness and the within-subject standard deviation of MUP velocities (Klaver-Król et al., 2007). The skewness expresses the relative proportions of faster and slower propagating MUPs. The standard deviation reflects the spread of MUP velocities (Klaver-Król et al., 2007; Lange et al., 2002).

In order to optimize the estimation of MUP velocities, a muscle must be activated in such a way that many MUPs are produced, and that the smaller MUPs are not obscured by larger ones. Therefore, we have developed a study design in the form of *a position task* at *low force levels* (Klaver-Król et al., 2007). In position tasks (as opposed to force tasks), a fixed inertial load is applied while the subject is required to maintain a target limb position (Hunter et al., 2002; Rudroff et al., 2005). Conversely, in force tasks, the limb position is fixed and the subject adjusts the effort to maintain a target force. A study on the biceps brachii has shown that position tasks result in a greater number of recruited motor units than force tasks (Mottram et al., 2005). A further study (this time involving an intrinsic hand muscle) has shown that position tasks render a greater amount of muscular activity (Maluf et al., 2005). In the present experiment, low-force contractions, performed as an isometric position task (traditionally referred to as static contractions), yielded a mixed pattern EMG which enabled many individual MUPs to be identified allowing a statistical analysis.

To date, as far as we know, surface EMG in sprinters and endurance athletes has only been assessed in leg muscles (Goswami et al., 2001; Rainoldi et al., 2008; Sadoyama et al., 1988). We wanted to know whether the distinguishing characteristics between such sportsmen could also be found in other muscles, reflecting inherited rather than trained properties. Consequently, a muscle was chosen that was little influenced by specific training. The biceps brachii (BB), arm muscle was selected since both long-distance runners and sprinters mainly train their leg muscles. The experiments were performed as position tasks, involving brief isometric contractions and low applied forces. We applied the force load to the palm as we intended to mimic natural circumstances, such as holding something in the hand. The features extracted from the EMG signal were the mean muscle conduction velocity (CV), the skewness of the MUP velocities (Sk), and the standard deviation of the MUP velocities (SD-mup). In order to strengthen confidence in the results, the CV was measured not only using the IPL method but also by the more commonly used cross-correlation (CC) method.

The goal of the present study has been to investigate whether the distribution of MUP velocities obtained during short, low-force contractions of the biceps brachii muscle is able to discriminate between sprinters and endurance athletes. Our expectation was that sprinters would show a higher mean CV and a greater proportion of faster propagating MUPs than endurance athletes.

2. Methods

2.1. Subjects

Fifteen competitive male sprinters and 18 competitive male endurance athletes volunteered for the experiment (age: sprinters 23 SD 4, endurance athletes 26 SD 5 years, $P = 0.052$; height: sprinters 182.5 SD 8.5, endurance athletes 187 SD 6.5 cm, $P = 0.12$; body mass: sprinters 77.0 SD 8.0, endurance athletes 74.5 SD 9.0 kg, $P = 0.47$). Endurance athletes tend on average to be slightly older than sprinters and we were not able to exactly match the groups in terms of ages. The athletes were recruited from their sports clubs by personal contact with the clubs' trainers. The protocol was conducted according to the Helsinki Declaration and approved by the local ethics committee. All participants gave their written informed consent. The inclusion criteria were (a) participation for at least 3 years in either sprint events (100–400 m running) or long-distance sports; (b) being well-trained, with a training frequency over the last year of at least three times a week and a total duration of at least three and a half hours per week. Exclusion criteria were drug abuse and practicing body building, because in body building the primary focus is on increasing muscle mass, and muscle fiber diameter influences the CV (Blijham et al., 2006).

The sprinters were all runners. The group of endurance athletes consisted of 14 distance runners, two marathon skaters, one triathlon athlete and one survival athlete. Two sprinters and one endurance athlete were left-handed. The experiments took place outside the competition period. Five days before the experiment, any unaccustomed exercise with the arms or strenuous exercise in general was not allowed. After running a marathon, a recovery period of at least 6 weeks was required.

2.2. Experimental set-up

The methods used have been described previously (Klaver-Król et al., 2007). Maximum voluntary contraction (MVC) of elbow flexors was measured at least 5 days before the experiment with a hand-held dynamometer (Lameris Instruments, Utrecht, The Netherlands). During the MVC measurements the subjects were sitting upright. The set-up is shown in Fig. 1A. The shoulder was slightly abducted and flexed at 45°; the elbow was firmly supported and flexed at 90°, while the forearm was supinated. The dynamometer was applied to the wrist. The force was kept for at least 3 s. The mean of three maximal values was taken as an MVC. The MVC was assessed at an elbow angle of 90° whereas the tests were performed at 135° (see below). The reason for not using an angle of 135° for the MVC assessments was that, when applying high forces at the wrist with this angle, subjects tended to overstretch the wrist and experienced pain, so influencing the results. This is not the case for a 90° angle. Further, 90° is the classic recommended position for MVC assessments (van der Ploeg et al., 1991) and previous work has shown that MVCs obtained at 135° and 90° elbow flexions are similar (Philippou et al., 2004).

During the experiment the subjects were sitting in a chair. The upper arm was slightly abducted and comfortably supported at 45° of the shoulder flexion; the forearm was free. The forearm was supinated. The set-up is shown in Fig. 1B. Subjects were asked to hold the forearm horizontally (elbow angle 135°). An individually adjustable horizontal bar was used showing the subjects the position at which the lower arm was really horizontal (i.e., when the elbow angle was 135°). Three levels of force were applied in blocks that were 3 min apart: unloaded, 10% and 20% of MVC. We used fixed sequences of force levels because our object was to investigate differences between sprinters and endurance athletes and not the effects of different force levels. In the loaded tests a bag filled with lead and sand was placed in the palm. A block consisted of three tests (three repetitions at the same force level) that were within a block separated 30 s from one another. The position was held for ~5 s; the measurements were performed during 3.8 s. During the tests, the load was applied to the palm, whereas the MVCs were assessed from the wrist (Fig. 1A and B). By applying

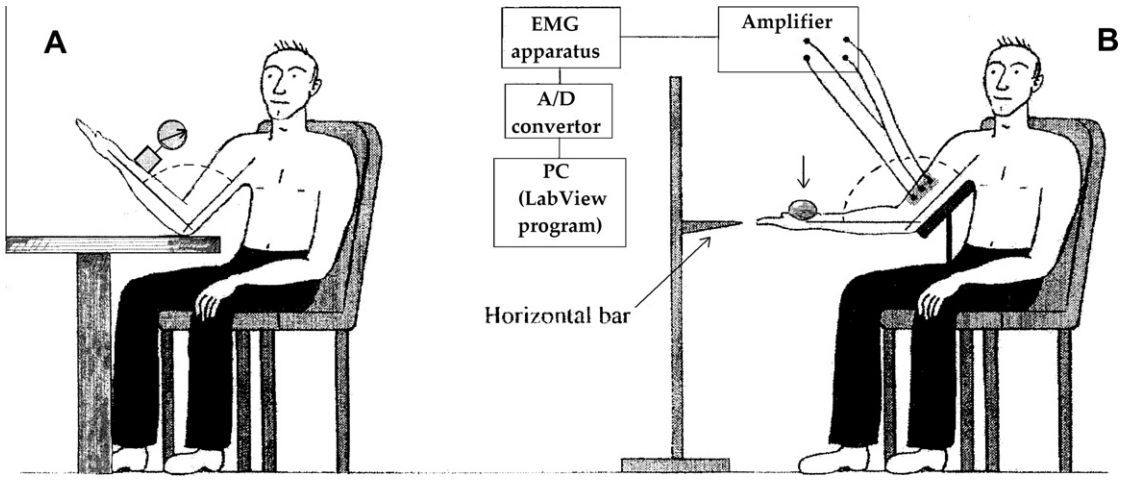

Fig. 1. Set-ups used for the assessment of the elbow flexors' strength with a hand-held dynamometer (A) and for the surface electromyographic (sEMG) measurements from the biceps brachii muscle (B). The strength is assessed using the classical method with an elbow angle of 90°, and a dynamometer applied to the wrist. However, sEMG measurements required a 135° elbow angle and the load applied to the palm. The maximum voluntary contractions (MVCs) obtained at the angles of 90° and 135° are known to be similar (Philippou et al., 2004). The difference between exerted load torques using wrist and palm force applications can be calculated from the ratio of the lengths from elbow to wrist and from elbow to palm.

the load to the palm, we were intending to mimic natural events, such as holding something in the hand. However, assessing the MVC from the palm was not feasible due to the weakness of the wrist's flexors relative to the elbow flexors, which would therefore influence the estimates. Distances measured from the elbow to the wrist and from the elbow to the palm enabled the true load torque exerted during the experiment to be calculated, and this was around 20–25% larger than the applied load.

2.3. EMG recording

Measurements were performed on the short head of the BB of the dominant arm. A surface electrode array consisted of three gold-coated electrodes in a bipolar configuration, with a common centre electrode. The electrodes were insulated in a rigid synthetic plate (Harwin, P25-3526). The diameter of the electrodes was 1.5 mm and the inter-electrode distance 10 mm. The skin was cleaned with 95% ethanol. The electrode array was placed parallel to the muscle fibers following a system we described previously: the proximal electrode was exactly on the distal one-quarter point of the BB, measured between the coracoid (proximal tendon insertion) and the crossing of the distal tendon with the elbow crease found by palpation (Klaver-Król et al., 2007). This position of the electrode array was about halfway between the endplate zone and the tendon, securing a sufficient distance from the endplate and the tendon (Masuda and Sadoyama, 1987; Sadoyama et al., 1985). Bipolar derivation was made from the proximal to distal direction. For the electrode position, a correlation coefficient (C-coef) of the sEMG signals was accepted at $r > 0.7$ (Hogrel et al., 1998; Merletti et al., 2001); the check was performed before and after the tests. The signals were differentially amplified (gain 2000–10,000) and band-pass filtered at 2–250 Hz by EMG apparatus (Viking IV, US). We used the relatively low cut-off frequency of 250 Hz to remove possible high frequency noise in the signal. This was done because we wanted well-defined peaks and was not interested in possible changes to the spectrum.

The skin temperature was measured with an electronic thermometer, the sensor was placed ~5 cm proximally from the derivation electrode. The average skin temperature was 32.2 SD 0.85 °C and was similar between the groups ($t(31) = 1.14$, $P = 0.26$). The skin thickness (measured with callipers after performing tests, on the spot where the electrodes were placed) did not differ between sprinters and endurance athletes (2.4 SD 0.8 mm and 2.6 SD 0.7 mm, respectively), $t(31) = 0.96$, $P = 0.34$.

2.4. Data processing

The signals were simultaneously A/D converted (sampling 10 kHz, 12 bits acquisition). Data were stored on a personal computer and analyzed with LabVIEW 6.1.

2.4.1. Determining the CV with the CC method (CV-cc)

We used the method as introduced by Nishizono et al. (1979). The C-coef check was performed over periods of 0.8 s. The CV calculations during the tests were over periods of 3.8 s.

2.4.2. The IPL method and its derived parameters

The signal was described by three parameters derived from MUP velocities (MUP-Vs = peak velocities): the muscle conduction velocity (CV-ipl), the skewness (Sk) and the standard deviation of MUP velocities (SD-mup). The MUP-Vs were obtained by the IPL method proposed by Lange et al. (2002) and modified by the authors (Klaver-Król et al., 2007).

2.4.3. Peak selection and estimation of peak velocities

The software was custom-designed and written in LabView. The negative MUP deviations in the signal were up-directed (Fig. 2). The procedure steps 1, 2 and 3 of the analysis were similar for the two sEMG signals. The steps were as follows:

(1) Finding the largest positive–negative amplitude difference in an epoch of 0.2 s. This amplitude difference is denoted as 100%.
(2) Finding declines. In a moving time window of ⩽4 ms (⩽40 sample points), declines (from negative to positive) are found of ⩾20% of the largest amplitude of an epoch of 0.2 s.
(3) Finding peaks. A peak is the highest (most negative) point previous to a decline.
(4) Finding pair-peaks in the second signal. A pair-peak is a peak in the second signal that satisfies the same requirements as those for the first signal's peak, and is found in a time window of 1.5–4.0 ms after the first signal's peak (this time interval corresponds with the CV values of 2.5–6.7 m s^{-1}) (Houtman et al., 2003; Lange et al., 2002).

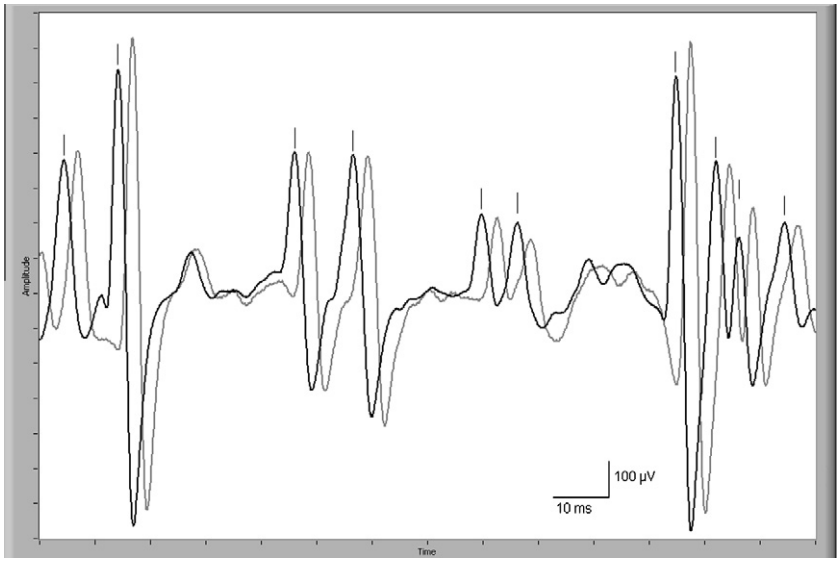

Fig. 2. Two differential electromyographic signals obtained from the biceps brachii muscle, with the subject's arm unloaded, the elbow flexed at 135° (position task). The negative peaks of the motor unit potentials have been identified using the inter-peak latency method (for clarity, only the peaks of the proximal signal are indicated). The large amount of motor unit potential activity produced despite the no load situation is attributed to the position task procedure.

(5) Excluding a double peak. If a peak from the first signal matches two different peaks from the second signal, then the second peak from the second signal is rejected.
(6) Estimating peak velocities. Peak velocities (MUP-Vs) are estimated for each pair of peaks from the latencies between the paired peaks and the inter-electrode distance.

The values of the three parameters (CV-ipl, Sk and SD-mup) were calculated from the gathering of all MUP-Vs obtained during the three tests (three repeated measurements of 3.8 s duration) at a given force level. The definitions of the parameters were:

CV-ipl: This was the average of the MUP-Vs elicited.
Sk: The within-subject skewness was the statistical measure of distribution of the MUP-Vs. The Sk indicated the relative proportion between the number of faster and slower propagating MUPs. When faster MUPs rise, and/or slower MUPs diminish, then the skewness becomes less positive. And reversely, when slower MUPs rise and/or faster ones fall, the skewness becomes more positive.
SD-mup: The within-subject standard deviation was the measure of spread of the MUP-Vs found.

2.5. Statistics

A repeated measures model of ANOVA was used (Kutner et al., 2005) that included within-subjects factor 'force' with three levels (unloaded, 10% MVC and 20% MVC) and between-subjects factor 'group' with two levels (sprinters and endurance athletes). If needed, an independent or paired samples *t*-test was applied. To evaluate association between variables, Pearson correlation coefficients were calculated. We used the SPSS program. A level of $P < 0.05$ was used to identify statistical significance.

3. Results

No association was found between the age of the subjects and the CV (for all tests and for both the CV-ipl and the CV-cc methods: $r < 0.178$). Nor did an association exist between the age and the skewness (for all tests: $r < 0.1$). The MVC was in sprinters 265.0 SD 29.0 N and in endurance athletes 242.5 SD 35.0 N, $P = 0.056$. There was no significant association between the MVC and the CV, or the MVC and skewness (CV for all tests and both methods: $r < 0.299$; skewness for all tests: $r < 0.296$).

3.1. Mean muscle conduction velocity (CV-ipl and CV-cc)

The CV obtained by the IPL method (CV-ipl) correlated positively with the CV by the CC method (CV-cc), (for all force levels: $r > 0.89$). Comparing the CV values of the two methods, the CV-ipl was at all force levels significantly lower than CV-cc, the difference being around 0.1–0.2 m s^{-1} (for all force levels: $t(32) > 5.65$, $P < 0.001$), see Fig. 3.

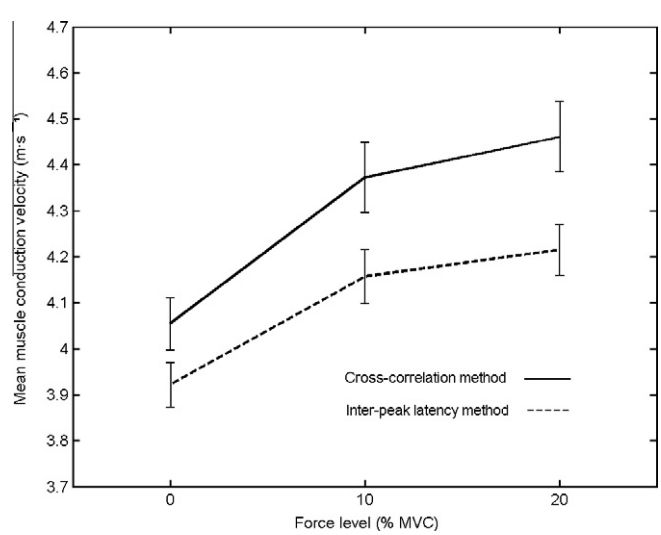

Fig. 3. Mean muscle conduction velocity (CV) as obtained using the inter-peak latency method (CV-ipl) and the cross-correlation method (CV-cc) in short static tests unloaded and loaded with 10% and 20% of maximum voluntary contraction force (MVC). Mean ± SE of the CV are given over 33 subjects. The CV-ipl is lower than the CV-cc.

Overall, augmenting forces raised the CV (main effect of force: for the CV-ipl $F(2,62) = 20.35$, $P < 0.001$; for the CV-cc $F(2,62) = 20.11$, $P < 0.001$). Both the CV-ipl and CV-cc were significantly higher in sprinters than in endurance athletes (between-subjects effects: CV-ipl $F(1,31) = 4.46$, $P = 0.043$; CV-cc $F(1,31) = 5.49$, $P = 0.026$), see Fig. 4a. The increases of the CV with forces were similar in sprinters and endurance athletes (force × group interaction: for IPL $F(2,62) = 0.08$, $P = 0.92$; for CC $F(2,62) = 0.56$, $P = 0.58$).

3.2. The within-subject skewness of MUP velocities (Sk)

Fig. 4b shows the averages of the Sk over 15 sprinters and 18 endurance athletes; the histograms in Fig. 4 present the distribution of MUP velocities in these subjects. Overall, with increasing forces the Sk became less positive, indicating that the relative proportion of faster propagating MUPs rose (main effect of force $F(2,60) = 3.43$, $P = 0.039$). In sprinters, the Sk was significantly less positive than in endurance athletes, which means that they had a greater relative proportion of faster propagating MUPs (between-subjects effects: $F(1,30) = 8.55$, $P = 0.007$). Histograms in Fig. 5 show this difference in the proportion of faster and slower propagating MUPs in sprinters versus endurance athletes, assuming the velocity of 4 m s^{-1} as an arbitrary threshold. Expressed in percentages, sprinters had 53%, 65% and 69% MUPs with velocities >4 m s^{-1} in the unloaded, 10% and 20% MVC tests, respectively; endurance athletes had 38%, 55% and 62%. Reaction to forces did not differ significantly between the groups (force × group interaction $F(2,60) = 2.49$, $P = 0.09$).

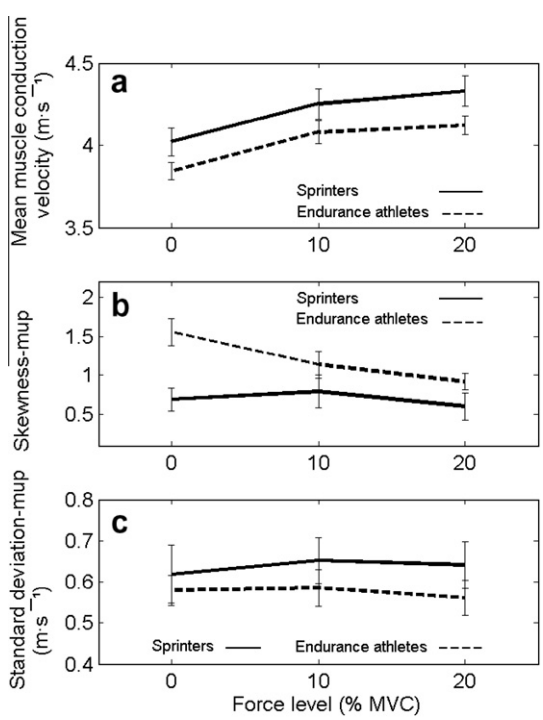

Fig. 4. Changes in surface EMG of sprinters and endurance athletes during short static tests at three force levels: unloaded, 10% and 20% of maximum voluntary contraction force (MVC). The changes are expressed by three parameters derived from the motor unit potential (MUP) velocities: the mean muscle conduction velocity (CV) (a); the skewness of MUP velocities (skewness-mup) (b); and the standard deviation of MUP velocities (standard deviation-mup) (c). Mean ± SE are shown over 15 sprinters and 18 endurance athletes. The CV of sprinters is higher (a) and skewness-mup is less positive (b) as compared with endurance athletes.

3.3. The within-subject standard deviation of MUP velocities (SD-mup)

The SD-mup values, and the reactions to forces were similar in both groups (between-subjects effect $F(1,31) = 1.05$, $P = 0.31$; force × group interaction $F(2,62) = 0.19$, $P = 0.83$), see Fig. 4c.

4. Discussion

The present study showed that distribution of MUP velocities in surface EMG is able to distinguish between the muscle properties of sprinters and of endurance athletes during short static low force contractions. Below, we will discuss the findings and consider certain aspects of the method.

4.1. Inter-peak latency versus cross-correlation in the CV measurements

The CVs obtained by the inter-peak latency method were highly correlated with the CVs found using the cross-correlation method, a finding in line with the study by Lange and colleagues (Lange et al. 2002). However, in the present study, the CV values found by the IPL were significantly lower than those using the CC (Fig. 3). This difference can be explained by the fact that a broader range of MUP velocities, including slower propagating MUPs, contributed to the average CV in the IPL. In contrast, in the CC measurements, mainly large, faster propagating MUPs determined the CV (Sollie et al. 1985).

4.2. Differences between sprinters and endurance athletes

Sprinters showed a higher mean muscle conduction velocity than endurance athletes and a larger proportion of faster propagating motor unit potentials. This is presented in Fig. 4a (the CV in sprinters is higher) and Fig. 4b (the skewness in sprinters is lower). Skewness is a relative distribution measure. However, the combination of a higher global CV, which is an absolute measure, with a greater proportion of faster propagating MUPs (lower skewness values) in sprinters shows where these 'higher' MUP velocities are really positioned on the absolute velocity scale. Such compound information about the absolute MUP velocities' distribution is presented in the form of histograms as shown in Fig. 5: sprinters compared with endurance athletes have a greater number of MUPs with velocities higher than 4 m s^{-1}, and/or smaller number with velocities lower than 4 m s^{-1} (arbitrary threshold).

Since type II fibers, belonging to large/fast MUs, propagate an action potential faster along a membrane than type I fibers, the conduction velocity of a given motor unit action potential will depend on the motor unit type (Andreassen and Arendt-Nielsen, 1987; Buchthal et al., 1973). Consequently, the mean conduction velocity and the skewness depend on the proportions of activated MU types in a muscle. The findings suggest that sprinters activate a greater number of larger/faster MUs than endurance athletes.

4.3. Position task and motor unit recruitment

Can faster/larger MUs be activated when using low forces of up to 20% MVC? Following the size principle, the larger/faster MUs have higher discharge thresholds, i.e., they are not recruited until high force levels are reached (Henneman et al., 1965). However, most of the studies to date on the recruitment order and recruitment gradation have employed force tasks (Gantchev et al., 1992; Henneman et al., 1965; De Luca et al., 1982; Kukulka and Clamann, 1981), whereas the present study was designed around a position task. Although the *recruitment order* of MUs is similar for both force tasks and position tasks (Tax et al., 1989), the *recruit-*

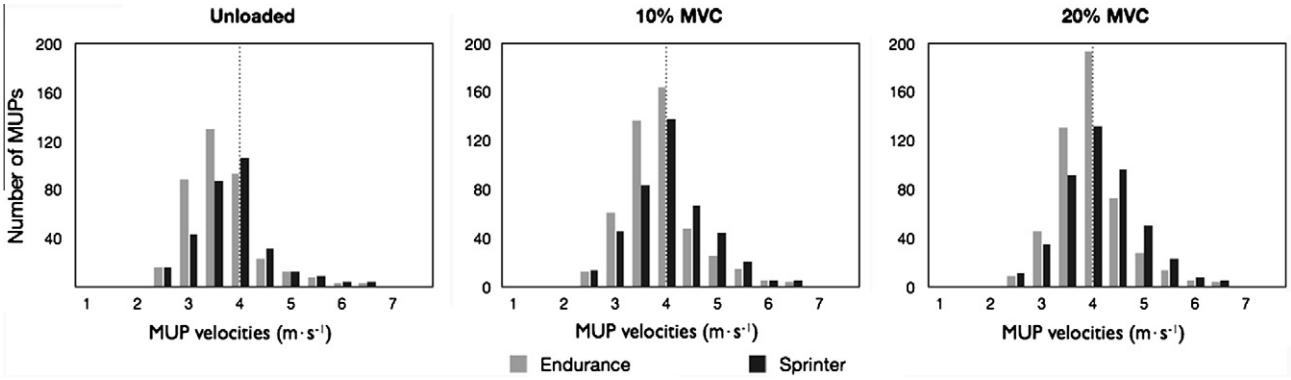

Fig. 5. Distribution of motor unit potential (MUP) velocities in sprinters and endurance athletes in short static tests at three force levels: unloaded, and loaded 10% and 20% of maximum voluntary contraction force (MVC). The results are expressed in terms of the mean number of MUPs with their propagation velocities over 15 sprinters (filled bars) and 18 endurance athletes (empty bars). Assuming an arbitrary threshold between faster and slower propagating MUPs of 4 m s^{-1}, sprinters compared with endurance athletes show a greater number of fast and/or a smaller number of slow propagating MUPs.

ment thresholds differ. In position tasks, the thresholds are lower (Tax et al., 1989) and the number of recruited MUs is greater (Mottram et al., 2005). Since MUs tend to discharge earlier in a position task, it seems possible that a proportion of the faster MUs, especially the intermediate MUs, will still be activated even when only low forces are involved. Thus, the greater proportion of higher MUP velocities in sprinters may indeed reflect a greater proportion of *intermediate* MUs recruited. An alternative explanation for the results is that, at the low force level used, only small MUs (with type I fibers) are recruited. The diameter of these type I fibers may be larger in sprinters and, if so, this will lead to higher propagation velocities along the membrane (and thus to higher MUP velocities) (Blijham et al., 2006).

4.4. Recruitment versus firing rate and velocity recovery function

Besides recruitment, the larger proportion of faster propagating MUPs/peaks (skewness) in sprinters could also be due to higher firing rates of the larger/faster propagating MUs. When the force increases, both the recruitment and firing rate of the activated MUs increase. However, in large muscles, such as biceps brachii, the force gradation relies more on recruitment than on firing rate (De Luca et al., 1982; Kukulka and Clamann, 1981). Therefore, especially at the low force levels used, recruitment will play the dominant role. Further, there is no task-dependent preference for firing rates in a position task (as is the case with recruitment): at forces lower than 20% MVC, the initial firing rates are similar in both types of tasks (MacGillis et al., 2003; Mottram et al., 2005).

In the present experiments, the firing rate was probably10–12 Hz, as assessed on the base of other studies on the biceps brachii muscle in position tasks (Mottram et al., 2005; Tax et al., 1989). The gain in the firing rate across the force range from unloaded to 20% MVC would be no more than 2–3 Hz (De Luca, 1985). Consequently, the firing rates, in the present study, must have contributed to the *total number* of MUPs produced and thus strengthened the distribution results. Nevertheless, it is unlikely that they had a significant influence on the results.

Increases in the proportion of faster MUPs/peaks with increasing force might be influenced by higher propagation velocities along muscle fiber membrane as a result of augmented firing rates (velocity recovery function of muscle fiber, or VRF) (Morimoto and Masuda, 1984; Nishizono et al., 1989). Farina and Falla (2008) have found that during contractions of the sternocleidomastoideus muscle, the conduction velocity of MUPs increases on average by 11% over the range of MU discharges from 7 Hz (minimum frequency) to 28 Hz (maximum frequency). Since, in the present study, the assessed gain in the firing rates over the force range from unloaded to 20% MVC was only 2–3 Hz, the increase in the MUP conduction velocity would be no more than 1–2%. The influence of the VRF on the present results thus seems negligible.

4.5. Genetics versus arm training

The between-group differences found may depend partly on inherited factors and partly on muscle training. During strength training of sufficient intensity and duration, the diameter of a muscle fiber increases (hypertrophy) (Bouchard et al., 1988; Brooks et al., 2005), and this may result in an increase in the fiber conduction velocity (Blijham et al., 2006; Nandedkar et al., 1985). However, the present investigations were performed on the arm muscles of sportsmen who in their sports especially use their legs. Further, the experiments were done outside of competition periods, and the subjects did no more than routine training exercises for their arms. Furthermore, an analysis of the data showed no relationship between CV and arm training intensity. Arm training might thus have been only a small factor in the results. Studies have shown that muscle fiber composition (and thus also fiber thickness) is chiefly determined by genetics (Komi, 1992; Thorstensson, 1976). Also, the effects of training are dictated by the genetic make-up of the muscles (Bouchard et al., 1988; Thomis et al., 1998). Consequently, although some influence of training cannot be excluded, the differences found between the sportsmen seem for the greater part to be intrinsic/inherited.

In conclusion, in a muscle that had not been specifically trained (biceps brachii), sprinters showed a higher mean CV and a greater proportion of faster propagating MUPs than endurance athletes. These findings may reflect intrinsic/inherited properties of the sportsmen. The results can be explained by the sprinters' earlier recruitment of intermediate motor units, which in the spectrum lie closer to the fast type of motor units. Alternatively, the recruited type I fibers/units may be thicker in sprinters, resulting in higher propagation velocities along the membrane. The physiological difference between the two types of sportsmen could still be seen during contractions at only low force levels, which is attributable to the use of a position task procedure. The inter-peak latency method, with its resulting MUP velocity distribution (skewness), appears valuable in distinguishing muscle properties. Findings based on the combination of mean CV and skewness of MUP velocities allow some extrapolation to the type of motor units.

Acknowledgments

We thank the members of the Department of Clinical Neurophysiology and the Department of Electronics of Ziekenhuis Groep Twente for their support in setting up the experiments. We are grateful to all our subjects for their freely given contribution to this project, and the trainers and leaders of the sports clubs for their support. Nizare R. Henriquez is currently at the Department of Clinical Neurophysiology, Institute of Neurology, University Medical Centre, Utrecht, The Netherlands. There are no conflicts of interest in the present study. The study was not sponsored.

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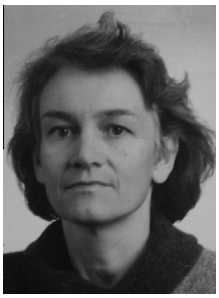

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